

The Algebra of Storytelling: A State Transition Benchmark for Spanning Multi-Entity Multi-Shot Narratives

Dongwoo Lee

Department of Computer Science and
Engineering
Sogang University
Seoul, Republic of Korea
dongwoolee@sogang.ac.kr

Yein Lee

Department of Computer Science and
Engineering
Sogang University
Seoul, Republic of Korea
leeyein@sogang.ac.kr

Jihoon Yang*

Department of Computer Science and
Engineering
Sogang University
Seoul, Republic of Korea
yangjh@sogang.ac.kr

Abstract

Multi-shot video generation models can now produce visually coherent sequences, yet often fail to execute prescribed entity entrances and exits, revealing a fundamental *consistency–dynamics trade-off*: mechanisms that preserve cross-shot appearance can resist the localized state changes required by dynamic narratives. Current evaluation paradigms emphasize visual quality and identity consistency, leaving entity-level narrative transitions undermeasured by reducing them to frame-wise similarities. We present the SPAN (State Progression and Algebraic Narrative) framework, which represents multi-entity, multi-shot scripts as state transition matrices encoding entity entrances, exits, and holds. We prove that two generating patterns (Relay and Split) span the canonical integer transition-matrix space under character role swapping and integer linear combination. This establishes representation-level algebraic completeness, rather than empirical enumeration of every physically realizable narrative. To evaluate generated videos, we introduce STEM (State Transition Error Matching). STEM constructs an error matrix by comparing observed transitions with the prescribed script, then applies a sparsity-aware decomposition that reduces sensitivity to static-cell detector noise, yielding *Prescribed Transition Match* (PTM) for transition fidelity and *Entity Presence Accuracy* (EPA) for cast compliance. On MEMS-1K, a basis-derived benchmark of 1,000 scenarios across seven interpretable patterns, PTM discriminates four representative generation systems and yields an inverse ordering relative to Identity Consistency. This ordering is preserved under multi-frame aggregation and an alternative VLM detector, while absolute scores remain detector-dependent. The error matrix further induces a four-type taxonomy of entity-transition failures, providing structured diagnostics complementary to visual-quality, identity-consistency, and human-preference evaluation. <https://merrybabyxmas.github.io/The-Algebra-of-Storytelling/>

CCS Concepts

• Computing methodologies → Computer vision; Neural networks.

*Corresponding author.



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Keywords

Multi-Shot Video Generation; State Transition Matrix; Entity-Level Narrative Evaluation; Video Benchmark

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1 Introduction

Consider a simple three-shot scene: a chef enters a kitchen, a waiter joins in the second shot, and the chef departs in the third. This elementary narrative requires precise *entity-level state control*—determining exactly who appears and disappears at each shot boundary. Yet state-of-the-art multi-shot video generators [2, 4, 12, 15, 28, 47] routinely fail at this task. The mechanisms designed to maintain cross-shot visual consistency—shared attention, in-context conditioning—can resist the localized changes needed for dynamic character entrances and exits, exposing a fundamental *consistency–dynamics trade-off*. We use this term for the inverse relationship observed between preserving visual coherence and executing prescribed entity-state changes.

Our scope is deliberately narrower than full storytelling. We evaluate whether named entities enter, exit, or remain present across shots, not actions, interactions, spatial relations, causal structure, camera motion, or emotional tone. Prominent video benchmarks [17, 23, 32] instead emphasize single-shot quality and cross-shot consistency through global embeddings or frame-level similarities. SPAN and STEM therefore complement, rather than replace, identity, visual-quality, text-alignment, and human-preference metrics.

The representation problem remains combinatorial: as shots and entities increase, presence configurations grow exponentially. Recent multi-shot benchmarks [18, 39, 45] evaluate extended horizons through empirically curated prompts, but do not formally characterize the transition representation space. This distinction matters because algebraic expressiveness, physical realizability, and empirical benchmark coverage are related but non-equivalent notions.

We reformulate entity-level narrative transitions through linear algebra. By abstracting a script into a *state transition matrix*, we prove that two generating patterns (Relay and Split) span the canonical integer matrix space under character role swapping and integer linear combination (**Theorem 3.4**). This representation-level result

provides a principled basis for constructing diagnostic patterns without claiming exhaustive empirical coverage. We contribute:

- The **SPAN framework** (§3): A formalization of entity-level multi-shot narratives as state transition matrices. We prove that two generating patterns span the canonical integer transition space (**Theorem 3.4**, §3.3) and extend the construction to arbitrary lengths and cast sizes (Theorem B.1, Appendix B).
- The **STEM evaluation pipeline** (§4): An automated diagnostic protocol built upon an algebraic *error matrix*. PTM and EPA separately measure prescribed entity-transition fidelity and entity-presence compliance while reducing sensitivity to static-cell detector noise.
- **MEMS-1K Benchmark** (§5): A basis-derived dataset of 1,000 scenarios covering seven interpretable transition topologies. We evaluate 4,000 videos from four representative systems and derive a **four-type failure taxonomy** for structured diagnosis.

2 Related Work

Text-to-Video Generation. Modern video synthesis rests on diffusion models [13, 30, 31], extended temporally via UNet attention [2, 12, 15], DiT architectures [4, 27, 36], flow-matching [21], and classifier-free guidance [14]. Open-source systems [9, 46] and space-time models [1, 8] have advanced visual fidelity, yet fine-grained multi-entity narrative control remains open.

Multi-Shot Video Generation. Multi-shot extensions address cross-shot coherence through a variety of structural paradigms. Current approaches utilize Consistent Self-Attention (CSA) [33, 47], cross-attention [24], latent initialization [25], LLM-guided layouts [20, 44], keyframe transitions [5, 40], flow-matching portraits [37], IP-Adapter [42], and in-context LoRA [6, 16]. Longer-horizon generation has also been explored through autoregressive token models [19, 35]. While these methods succeed in maintaining subject identity, their diverse temporal conditioning mechanisms result in varying capacities to handle character transitions, such as appearances and disappearances. In this work, we evaluate representatives from four distinct paradigms—CSA (StoryDiffusion), flow-matching (EchoShot), in-context DiT (VIC), and cascaded T2I+I2V (VGoT)—providing a cross-paradigm diagnostic comparison focused explicitly on entity-level narrative dynamics.

Video Generation Benchmarks. Existing benchmarks quantify generation quality via embedding similarities (Table 1). FID [11] and FVD [34] measure distributional fidelity but are agnostic to narrative structure; CLIPScore [10] captures text–image alignment at the single-frame level; VBench [17, 43] and EvalCrafter [23] compute frame-level CLIP [29] and DINO [26] based similarities that conflate subjects with backgrounds; T2V-CompBench [32] targets compositional ability but is restricted to single clips. Recent multi-shot benchmarks—ShotAdapter [18], MovieBench [39], LoCoT2V [45]—begin to address cross-shot coherence but lack explicit metrics for multi-entity narrative dynamics and rely on empirically curated prompt sets without formal coverage guarantees. SPAN complements heuristic sampling with an algebraic characterization of the entity-transition representation space.

Table 1: Benchmark comparison. ✓ supported, ✗ not, △ partial. “Alg. Span” denotes a formal characterization of the transition representation space, not exhaustive empirical coverage.

Benchmark	Multi-shot	Entity Level	Absence Check	Alg. Span	Error Tax.
VBench 2.0 [43]	✗	✗	✗	✗	✗
T2V-CompBench [32]	✗	△	✗	✗	✗
MovieBench [39]	✓	△	✗	✗	✗
LoCoT2V [45]	✓	✗	✗	✗	✗
ShotAdapter [18]	✓	△	✗	✗	✗
SPAN + STEM (Ours)	✓	✓	✓	✓	✓

3 The SPAN Framework

3.1 Narratives as State Transition Matrices

A multi-shot video consists of T consecutive shots depicting up to N named entities (e.g., “a tuxedo cat,” “an orange tabby”). To mathematically formalize their appearances, we define the binary *presence matrix* $S \in \{0, 1\}^{T \times N}$, which records entity visibility across shots: $S_{t,e} = 1$ if entity e appears in shot t , and 0 otherwise.

To evaluate the dynamic storytelling (entrances and exits), we extract the narrative derivative via temporal differencing, yielding the *state transition matrix* ΔS :

$$\Delta S = S_{2:T} - S_{1:T-1}, \quad \Delta S_{t,e} \in \{-1, 0, +1\}, \quad (1)$$

where the subtraction is performed element-wise across all N entity columns, and the values algebraically encode the narrative operations: $+1$ (*enter*), -1 (*exit*), and 0 (*hold*). For example, consider a “Relay” narrative ($A \rightarrow AB \rightarrow B$), where entity B enters at transition 1, and entity A exits at transition 2:

$$S = \begin{matrix} & \text{Entity } A & \text{Entity } B \\ \text{Shot 1} & \begin{bmatrix} 1 & 0 \end{bmatrix} \\ \text{Shot 2} & \begin{bmatrix} 1 & 1 \end{bmatrix} \\ \text{Shot 3} & \begin{bmatrix} 0 & 1 \end{bmatrix} \end{matrix} \quad \Delta S = \begin{matrix} & \text{Entity } A & \text{Entity } B \\ \text{Shot 1} \rightarrow 2 & \begin{bmatrix} 0 & +1 \end{bmatrix} \\ \text{Shot 2} \rightarrow 3 & \begin{bmatrix} -1 & 0 \end{bmatrix} \end{matrix} \quad (2)$$

In these matrices, rows correspond to temporal transitions ($t = 1$: shot 1→2; $t = 2$: shot 2→3) and columns correspond to distinct entities (e.g., A, B).

Canonical Embedding. For standard 3-shot narratives ($T = 3$), ΔS inherently has 2 rows representing transitions. To unify the algebraic treatment across varying entity counts (up to $N = 3$), we embed all matrices into a canonical form via spatial zero-padding. For instance, the 2×2 Relay matrix extends naturally into $\mathbb{Z}^{2 \times 3}$ (the space of all 2×3 integer matrices):

$$\underbrace{\begin{bmatrix} 0 & +1 \\ -1 & 0 \end{bmatrix}}_{\text{Relay } (2 \times 2)} \rightarrow \underbrace{\begin{bmatrix} 0 & +1 & 0 \\ -1 & 0 & 0 \end{bmatrix}}_{\text{Canonical } (2 \times 3)}. \quad (3)$$

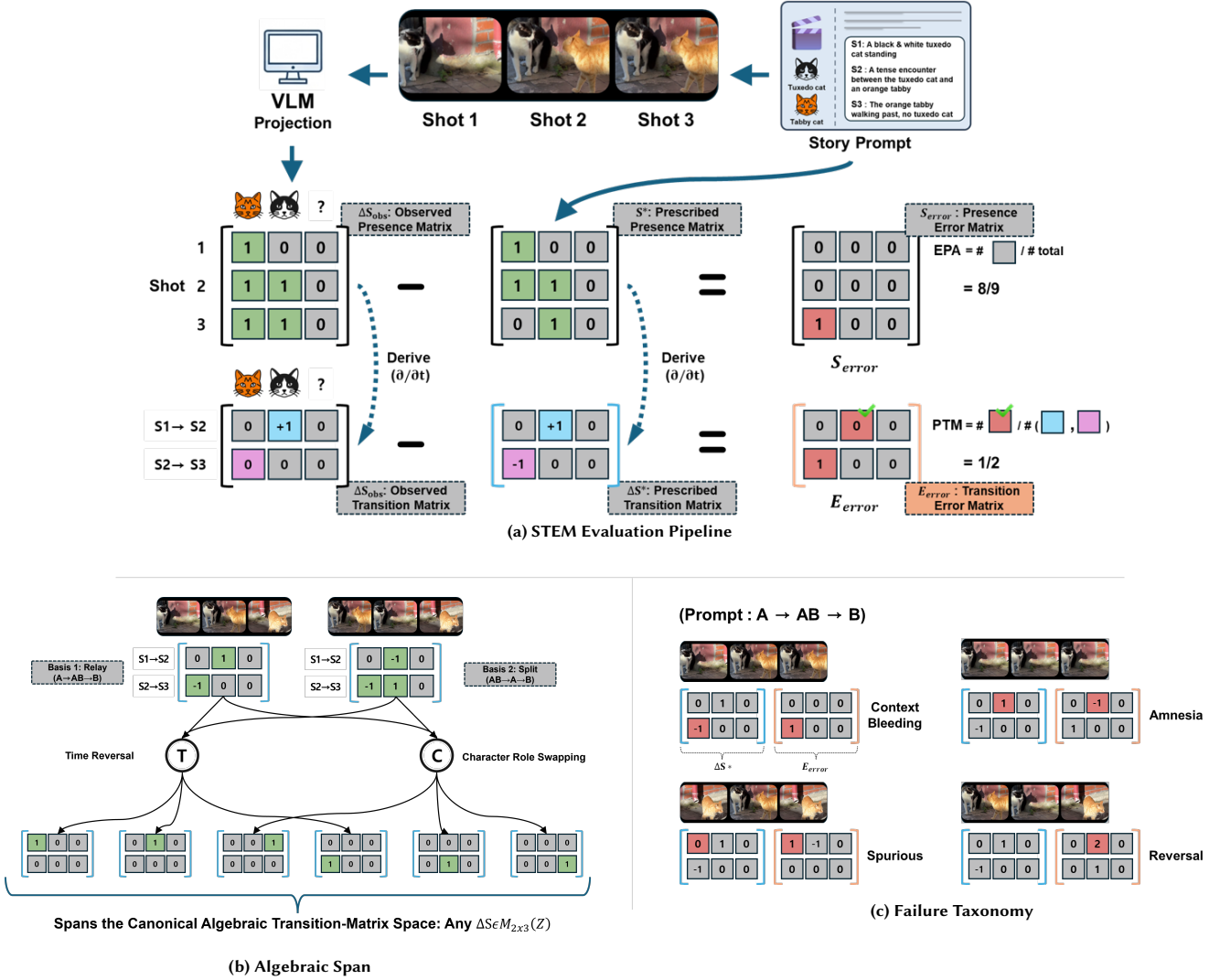


Figure 1: SPAN framework overview. source frames by Zeynep Coşkun via Pexels, used under the Pexels License. (a) The STEM pipeline: a VLM projects each shot into a binary presence vector, forming S_{obs} . Temporal differencing yields ΔS_{obs} , and the error matrix $E = \Delta S_{obs} - \Delta S^*$ diagnoses failures and feeds PTM/EPA metrics. (b) Algebraic span: two generating matrices ($\mathcal{B}_1$: Relay, $\mathcal{B}_2$: Split) produce all single-entity matrices J_{ij} and span the canonical space $\mathbb{Z}^{2 \times 3}$; this representation-level result is distinct from empirical benchmark coverage. (c) Four failure types: Context Bleeding (Type I), Amnesia (Type II), Spurious (Type III), and Reversal (Type IV).

This canonical projection enables a unified completeness proof over a single matrix space. The theoretical extension to arbitrary T and N is provided in Appendix B.

3.2 Linear Operators and the Narrative Basis

To systematically span the narrative space, we first define two linear operators that manipulate narrative topologies, followed by the fundamental basis matrices.

Definition 3.1 (Character Role Swapping). Let σ_{ij} denote the transposition swapping columns i and j . This operator reassigns entity roles without altering the underlying sequence of entrances and

exits:

$$(\sigma_{ij} \circ \Delta S)_{t,e} = \Delta S_{t, \sigma_{ij}^{-1}(e)}. \quad (4)$$

Compositions read right-to-left: $\sigma_{23} \circ \sigma_{12}$ first swaps columns 1 and 2, then 2 and 3. For instance, applying σ_{12} to Split ($AB \rightarrow A \rightarrow B$) exchanges the entity roles and yields the swap-over topology $AB \rightarrow B \rightarrow A$:

$$\sigma_{12} \circ \underbrace{\begin{bmatrix} 0 & -1 & 0 \\ -1 & +1 & 0 \end{bmatrix}}_{\text{Split } (AB \rightarrow A \rightarrow B)} = \underbrace{\begin{bmatrix} -1 & 0 & 0 \\ +1 & -1 & 0 \end{bmatrix}}_{\text{Swap-Over } (AB \rightarrow B \rightarrow A)}.$$

Its presence sequence is $S' = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}$: the first transition removes A , and the second reintroduces A while removing B . Thus, the final appearance of A is prescribed by the transformed topology rather than introduced accidentally.

Definition 3.2 (Time Reversal). The time reversal operator $\mathcal{T}$ rewinds the narrative by inverting the temporal order and negating all signs, thereby converting entries to exits and vice versa:

$$(\mathcal{T} \circ \Delta S)_{t,e} = -\Delta S_{T-t,e}. \quad (5)$$

Algebraically, this flips the matrix vertically and inverts all signs. This operator fundamentally transforms the narrative structure by reversing the directional flow of the story. For instance, applying $\mathcal{T}$ to the Split matrix ($AB \rightarrow A \rightarrow B$) rewinds the separation event, transforming it into a narrative where the entities join together ($B \rightarrow A \rightarrow AB$), yielding a Convergence pattern:

$$\mathcal{T} \circ \underbrace{\begin{bmatrix} 0 & -1 & 0 \\ -1 & +1 & 0 \end{bmatrix}}_{\text{Split } (AB \rightarrow A \rightarrow B)} = \underbrace{\begin{bmatrix} +1 & -1 & 0 \\ 0 & +1 & 0 \end{bmatrix}}_{\text{Convergence } (B \rightarrow A \rightarrow AB)}.$$

Definition 3.3 (Topological Basis Matrices). We identify two matrices that serve as the fundamental algebraic basis of the narrative space (represented in canonical 2×3 form):

$$\underbrace{\begin{bmatrix} 0 & +1 & 0 \\ -1 & 0 & 0 \end{bmatrix}}_{\text{Relay } (\mathcal{B}_1)}, \quad \underbrace{\begin{bmatrix} 0 & -1 & 0 \\ -1 & +1 & 0 \end{bmatrix}}_{\text{Split } (\mathcal{B}_2)}. \quad (6)$$

- **Relay** ($\mathcal{B}_1$): A sequential hand-off ($A \rightarrow AB \rightarrow B$).
- **Split** ($\mathcal{B}_2$): Co-occurring entities separate ($AB \rightarrow A \rightarrow B$).

3.3 Completeness Theorem and Derived Patterns

We now prove that these generating patterns are sufficient to span the canonical integer transition-matrix space. To formalize this, let $J_{ij} \in \mathbb{Z}^{2 \times 3}$ denote the single-entry matrix with $+1$ at position (i, j) and 0 elsewhere (e.g., $J_{12} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$).

THEOREM 3.4 (COMPLETENESS OF THE CANONICAL MATRIX SPACE). Under character role swapping ($\sigma \in \Sigma_3$) and integer-coefficient linear combination, the two basis patterns $\mathcal{B}_1$ (Relay) and $\mathcal{B}_2$ (Split) span the entire state transition matrix space $\mathbb{Z}^{2 \times 3}$.

PROOF. To prove completeness, it is sufficient to construct all six single-entry matrices $\{J_{ij}\}$ from $\mathcal{B}_1$ and $\mathcal{B}_2$, as any integer matrix is inherently a $\mathbb{Z}$ -linear combination of this standard basis.

Step 1: Isolate row-difference $J_{11} - J_{21}$. Consider the linear combination of $\mathcal{B}_1$ and its permuted variants:

$$\begin{aligned} & \mathcal{B}_1 - (\sigma_{13} \circ \mathcal{B}_1) + (\sigma_{23} \circ \sigma_{12} \circ \mathcal{B}_1) \\ &= \begin{bmatrix} 0 & 1 & 0 \\ -1 & 0 & 0 \end{bmatrix} - \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & -1 \end{bmatrix} + \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & -1 \end{bmatrix} = J_{11} - J_{21}. \end{aligned} \quad (7)$$

Step 2: Extract the standard basis $\{J_{ij}\}$. By adding a permuted Split matrix $\sigma_{12} \circ \mathcal{B}_2 = \begin{bmatrix} -1 & 0 & 0 \\ 1 & -1 & 0 \end{bmatrix}$ to Eq. (7), we isolate $-J_{22}$:

$$\begin{aligned} & (J_{11} - J_{21}) + \sigma_{12} \circ \mathcal{B}_2 \\ &= \begin{bmatrix} 1 & 0 & 0 \\ -1 & 0 & 0 \end{bmatrix} + \begin{bmatrix} -1 & 0 & 0 \\ 1 & -1 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & -1 & 0 \end{bmatrix} = -J_{22}. \end{aligned} \quad (8)$$

The remaining elementary matrices are subsequently derived via:

$$\begin{cases} J_{22} = -1 \cdot (-J_{22}), & \{J_{21}, J_{23}\} = \{\sigma_{12}, \sigma_{23}\} \circ J_{22} \\ J_{11} = (J_{11} - J_{21}) + J_{21}, & \{J_{12}, J_{13}\} = \{\sigma_{12}, \sigma_{13}\} \circ J_{11} \end{cases} \quad (9)$$

where all six components $\{J_{ij}\}$ are generated by the basis:

$$\{J_{11}, J_{12}, J_{13}, J_{21}, J_{22}, J_{23}\} \subset \text{span}_{\mathbb{Z}}(\mathcal{B}_1, \mathcal{B}_2, \Sigma_3). \quad (10)$$

□

Appendix A proves that $\mathcal{B}_2$ is algebraically essential, as all permutations of $\mathcal{B}_1$ share an invariant row-sum constraint ($R_1 + R_2 = 0$).

Constructing Derived Narrative Patterns. With the Completeness Theorem characterizing the canonical algebraic space, we utilize the generating patterns $(\mathcal{B}_1, \mathcal{B}_2)$, the defined operators ($\sigma, \mathcal{T}$), and the extracted single-entry matrices (J_{ij}) to derive five complex benchmark topologies:

- **Reverse Relay** ($B \rightarrow AB \rightarrow A$): $\sigma_{12} \circ \mathcal{B}_1 = \mathcal{T} \circ \mathcal{B}_1 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \end{bmatrix}$.
- **Convergence** ($A \rightarrow B \rightarrow AB$): $\mathcal{T} \circ (\sigma_{12} \circ \mathcal{B}_2) = \begin{bmatrix} -1 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}$.
- **Accumulation** ($A \rightarrow AB \rightarrow ABC$): $J_{12} + J_{23} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$.
- **Reduction** ($ABC \rightarrow AB \rightarrow A$): $\mathcal{T} \circ (\text{Accumulation}) = \begin{bmatrix} 0 & 0 & -1 \\ 0 & -1 & 0 \end{bmatrix}$.
- **Sliding Window** ($AB \rightarrow BC \rightarrow C$): $-J_{11} + J_{13} - J_{22} = \begin{bmatrix} -1 & 0 & 1 \\ 0 & -1 & 0 \end{bmatrix}$.

3.4 Theoretical & Practical Implications

Algebraic Span vs. Empirical Coverage. Theorem 3.4 is a representation level statement: $\mathcal{B}_1$ and $\mathcal{B}_2$ span $\mathbb{Z}^{2 \times 3}$ under permutations and integer linear combination. Restricting entries to $\{-1, 0, +1\}$ yields $3^6 = 729$ algebraically admissible matrices, but binary presence sequences allow only seven transition pairs per entity, giving $7^3 = 343$ physically realizable matrices. MEMS-1K does not enumerate all 343; its seven patterns are canonical, interpretable samples derived from the algebraic generators. Accordingly, algebraic span, physical feasibility, and empirical benchmark coverage should not be conflated.

Scalability. For T -shot, N -entity videos, $\mathbb{Z}^{(T-1) \times N}$ can be represented by zero-padding and temporally shifting the generating patterns (Theorem B.1, Appendix B). This establishes representational extensibility, while longer-horizon empirical validation remains a separate question.

4 The STEM Evaluation Methodology

We introduce STEM (State Transition Error Matching), a fully automated pipeline that bridges the continuous visual space and our discrete algebraic framework by employing VLMs as state detectors.

4.1 VLM-Based State Detection

Evaluating a generated video requires projecting its continuous visual space into the discrete observed presence matrix $S_{\text{obs}} \in \{0, 1\}^{T \times N}$. We accomplish this by extracting the middle frame of each shot $t \in \{1, \dots, T\}$ and querying a VLM with a binary question for each prescribed entity e :

$$S_{\text{obs}}[t, e] = \mathbb{I}[\text{VLM}(\text{frame}_t, \text{"Is entity } e \text{ visible?"}) = \text{Yes}]. \quad (11)$$

We use the *middle* frame by default because it is temporally separated from shot-boundary artifacts such as fades and cuts. This

protocol measures *inter-shot* presence states, not within-shot temporal persistence. To test sampling sensitivity, we replace middle-frame detection with 3-frame and 5-frame majority voting. PTM changes from (.243, .167, .094, .070) to (.225, .160, .090, .044) and (.207, .161, .087, .035) for (SD, Echo, VGoT, VIC), respectively, while both alternatives preserve the ordering SD > Echo > VGoT > VIC; EPA is likewise stable (Appendix Y). We use Qwen2-VL-7B-Instruct [38] as the primary VLM and query each prescribed entity independently with a binary question, producing the discrete states required by the matrix evaluation.

4.2 Algebraic Error Matrix and Failure Taxonomy

Given S_{obs} , we compute the observed state transition matrix $\Delta S_{\text{obs}} = S_{\text{obs}, 2:T} - S_{\text{obs}, 1:T-1}$. The cornerstone of our diagnostic pipeline is the *error matrix*, which captures all narrative deviations within a single algebraic object.

Definition 4.1 (Error Matrix). Let ΔS^* be the prescribed transition matrix. The error matrix $E \in \{-2, -1, 0, +1, +2\}^{(T-1) \times N}$ is defined as:

$$E = \Delta S_{\text{obs}} - \Delta S^*. \quad (12)$$

A perfect narrative execution yields $E = \mathbf{0}$. Any non-zero entry uniquely localizes a narrative failure to a specific transition–entity pair (t, e) .

The algebraic structure of E naturally induces an exhaustive taxonomy of narrative failures (Fig. 1(c)).

Definition 4.2 (Failure Taxonomy). Based on the joint state of $(E_{t,e}, \Delta S_{t,e}^*)$, we classify narrative failures into four distinct types, enabling fine-grained error localization:

- **Type I (Context Bleeding):** $\Delta S_{t,e}^* = -1 \wedge E_{t,e} > 0$ (entity persists despite prescribed exit).
- **Type II (Amnesia):** $\Delta S_{t,e}^* = +1 \wedge E_{t,e} < 0$ (entity fails to appear at prescribed entry).
- **Type III (Spurious):** $\Delta S_{t,e}^* = 0 \wedge E_{t,e} \neq 0$ (unintended transition during hold phase).
- **Type IV (Reversal):** $|E_{t,e}| = 2$ (transition diametrically opposite to prescription).

Note. Type IV represents the most severe compounded failure and is algebraically a strict subset of Types I and II: a single $|E_{t,e}| = 2$ entry simultaneously satisfies the condition for its base type (I or II) and for Type IV. The failure counts in Table 6 report each type independently.

4.3 Sparsity-Aware Metrics

A naive aggregation of E would compute its Frobenius norm $\|E\|_F$. However, because hold cells ($\Delta S_{t,e}^* = 0$) empirically outnumber active transitions by a factor of 2 to 4 in standard narratives, VLM detection noise in these static regions disproportionately inflates the unnormalized distance. This effectively drowns the true narrative signal ($\|E\|_F$ achieves only 1 of 6 pairwise significances at $\alpha = 0.05$; see Appendix R). To rigorously isolate narrative fidelity, we decompose the evaluation into two complementary, sparsity-aware metrics.

Definition 4.3 (Prescribed Transition Match, PTM). PTM quantifies *microscopic transition fidelity* by evaluating only the prescribed dynamic changes. Let $\mathcal{M} \in \{0, 1\}^{(T-1) \times N}$ be the binary prescription mask where $\mathcal{M}_{t,e} = \mathbb{1}[\Delta S_{t,e}^* \neq 0]$. Then:

$$\text{PTM} = 1 - \frac{\|E \odot \mathcal{M}\|_0}{\|\mathcal{M}\|_0} \quad (13)$$

where $\odot$ denotes the Hadamard product, $\|\cdot\|_0$ counts non-zero entries, and $\|\mathcal{M}\|_0$ is the total number of prescribed transitions.

By masking out hold cells, PTM reduces the dominant contribution of static-region VLM noise while measuring the prescribed entity transitions.

Definition 4.4 (Entity Presence Accuracy, EPA). EPA assesses *macroscopic cast compliance* by computing the element-wise accuracy of the underlying presence states:

$$\text{EPA} = 1 - \frac{\|S_{\text{obs}} - S^*\|_0}{T \cdot N} \quad (14)$$

Together, PTM (“Did the right *transitions* happen?”) and EPA (“Are the right *entities* present?”) provide a mathematically decoupled, sparsity-aware evaluation of multi-shot video generation—counting discrete mismatches via ℓ_0 norms in complementary matrix subspaces.

5 The MEMS-1K Benchmark

We construct **MEMS-1K (Multi-Entity Multi-Shot 1K)**, a dataset of 1,000 scenarios (3,000 shots) covering seven canonical, basis-derived narrative patterns rather than all physically realizable matrices. Relay, Split, and Accumulation receive 200 scenarios each; the remaining patterns receive 100 each (Table 3). Reversed pairs are retained because the evaluated systems exhibit marked directional differences between entrances and exits (Appendix L).

Narrative Pattern Design. Opposite pairs such as Relay/Reverse Relay and Accumulation/Reduction share closely related algebraic structure but reverse the direction of entity-state changes. Their comparison separates general difficulty in modifying a scene from difficulty specifically associated with removing an established entity. The seven selected topologies thus provide interpretable diagnostic coverage, not exhaustive sampling of the 343 physically realizable three-shot, three-entity transition matrices.

Structured Prompt Generation. We systematically generate scenarios using Gemini 1.5 [7] through strict structural templates. Each scenario first defines a target presence matrix S^* , with $S_{t,e}^* = 1$ iff entity e should appear in shot t ; per-shot prompts are instantiated from this symbolic specification and automatically checked against the target entity set. Templates select entities from Human, Animal, and Occupation categories and scenes from 10 environments. Each scenario has a cinematic `gen_prompt` for generation and a concise `eval_prompt` for VLM detection, decoupling generation wording from evaluation queries.

Semantic Diversity & Disentanglement. The entities span Human, Animal, and Occupation categories, while scenes cover ten domestic, urban, nature, and professional environments. Unique entity–scene combinations reduce surface-level shortcuts. To limit ambiguity in independent VLM queries (e.g., “dog” vs. “wolf”), we filter entity

Table 2: Main results. Best in bold. PTM = Prescribed Transition Match (primary). EPA = Entity Presence Accuracy (secondary). PTM 95% CIs from $B = 10,000$ bootstrap. Per-pattern PTM shown for all 7 patterns (B = basis, D = derived). Detailed per-pattern EPA scores are provided in Appendix I.

Model	Overall		95% CI		PTM by Pattern						
	PTM $\uparrow$	EPA $\uparrow$	PTM low	PTM high	<i>Relay^B</i>	<i>Split^B</i>	<i>Accum.^D</i>	<i>Conv.^D</i>	<i>Sl. Win.^D</i>	<i>Red.^D</i>	<i>Rev. Rel.^D</i>
SD	.243	.593	.226	.262	.190	.233	.253	.277	.357	.250	.200
Echo	.167	.563	.151	.183	.138	.165	.095	.187	.270	.195	.225
VGoT	.094	.554	.082	.107	.085	.108	.070	.053	.103	.110	.150
VIC	.070	.546	.060	.082	.080	.048	.072	.067	.077	.080	.080

Table 3: Distribution of narrative patterns in MEMS-1K. *Trans.* denotes the number of non-zero entries (active state changes) in the prescribed transition matrix ΔS^* . *Relay*, *Split*, and *Accumulation* are densely sampled ($n=200$): the first two as algebraic bases, and *Accumulation* as the only pattern that uniquely introduces a third entity.

Pattern	Shot Structure	Scenarios	Trans.	Type
Relay	$A \rightarrow AB \rightarrow B$	200	2	Basis
Split	$AB \rightarrow A \rightarrow B$	200	3	Basis
Accumulation	$A \rightarrow AB \rightarrow ABC$	200	2	Derived
Convergence	$A \rightarrow B \rightarrow AB$	100	3	Derived
Sliding Window	$AB \rightarrow BC \rightarrow C$	100	3	Derived
Reduction	$ABC \rightarrow AB \rightarrow A$	100	2	Derived
Reverse Relay	$B \rightarrow AB \rightarrow A$	100	2	Derived

pairs to CLIP text similarity ≤ 0.45 ; the retained mean is 0.23 ($\sigma = 0.08$). This threshold is a detector-disambiguation trade-off, not a claim about realistic casts, and may underrepresent narratives containing visually or semantically similar characters.

6 Experiments

6.1 Experimental Setup

Models Evaluated. We evaluate four systems representing distinct generation paradigms: (1) **Consistent Self-Attention** (Story-Diffusion [47]; related to Consistory [33]), (2) **Flow-matching conditioning** (EchoShot [37]; related to Wan [36] and AnimateDiff [9]), (3) **In-context DiT** (VIC [6]; CogVideoX-5B [41] + LoRA [16]), and (4) **Cascaded T2I $\rightarrow$ I2V** (VGoT [44]; related to MEVG [25]). These systems provide a representative cross-paradigm diagnostic comparison, not an exhaustive leaderboard; architecture-level interpretations are therefore hypotheses supported by the evaluated representatives. We abbreviate them as **SD**, **Echo**, **VGoT**, and **VIC**. Generations use fixed random seeds.

Evaluation Protocol. For each generated 3-shot video, we extract the middle frame of each shot and query Qwen2-VL-7B-Instruct [38] using binary entity presence questions (Eq. (11)). All 4,000 generated videos (1,000 per model) are processed across 4 $\times$ RTX 4090 GPUs.

Table 4: Model architectures and configurations.

Model	Paradigm	Backbone	Steps
SD	CSA + I2V	SDXL+SEINE	30+50
Echo	Flow-match	Wan2.1-1.3B	50
VGoT	T2I $\rightarrow$ I2V	Kolors+DC	25+50
VIC	In-context	CogVideoX-5B	50

Table 5: Pairwise significance (PTM). p -values (Mann-Whitney U), Cohen’s d .

	SD	Echo	VGoT	VIC
SD	—	3.3×10^{-10}	6.0×10^{-38}	2.7×10^{-53}
Echo	$d=0.28$	—	2.3×10^{-11}	4.0×10^{-21}
VGoT	$d=0.60$	$d=0.32$	—	2.9×10^{-3}
VIC	$d=0.72$	$d=0.44$	$d=0.13$	—

6.2 Main Results

Table 2 demonstrates the diagnostic power of the STEM evaluation pipeline.

Finding 1: Rigorous Method Discrimination. All pairwise PTM comparisons are statistically significant (Mann-Whitney U , $p < 0.005$, Table 5), and the 95% bootstrap confidence intervals show clear separation (Table 2). The Kruskal-Wallis test further yields $H = 302.7$ ($p < 10^{-65}$), providing strong evidence of performance differences among the four evaluated systems under MEMS-1K.

Finding 2: Consistent Rank Reversal Against IC. PTM and EPA produce the same ordering (SD > Echo > VGoT > VIC), which is reversed relative to reported Identity Consistency (IC) scores [18]: VIC 0.94, VGoT 0.88, Echo 0.84, and SD 0.81. Within these four systems, VIC has the highest IC and lowest PTM, whereas SD has the lowest IC and highest PTM, yielding $\rho_s = -1.0$. This evaluated-set association motivates the *consistency-dynamics trade-off* analysis in §6.5; it does not establish a universal causal law.

Finding 3: SD Leads Across Topologies. SD achieves the highest PTM in 6 of the 7 narrative patterns. Its advantage is particularly pronounced on *Sliding Window* (0.357 vs. 0.270 for Echo), which is the most complex pattern involving all 3 entities across both state transitions.

6.3 Metric Sensitivity Across Patterns

The per-pattern breakdown (Table 2) reveals how STEM exposes architectural limitations that remain invisible to aggregate scalar scores.

Pattern Difficulty Gradient. The PTM spread (best minus worst system) ranges from 0.11 on Relay to approximately 0.28 on Sliding Window. Patterns containing three prescribed active transition cells (Split, Convergence, Sliding Window) generally provide stronger discrimination than those containing two, indicating that denser entity-state changes better expose capability gaps.

Exposing Architectural Bottlenecks. SD performs strongly on dense multi-entity transitions (e.g., Convergence: 0.277), whereas VGoT reaches 0.053 on the same pattern. This result is consistent with a possible planning–rendering bottleneck when entities absent from an initial layout must be introduced later by DynamiCrafter [40]. VIC varies little across patterns (PTM range ≈ 0.032), suggesting that strong in-context conditioning may constrain entity-state flexibility in the evaluated setting.

6.4 Error Taxonomy Analysis

Fewer than 1.5% of videos are entirely failure-free, even for the best-performing model (Table 6). This stark statistic underscores that multi-entity narrative execution remains an unsolved challenge. STEM’s error taxonomy moves beyond this aggregate failure rate to provide actionable architectural diagnostics (Fig. 2).

Context Bleeding (Type I) in In-Context Models. VIC has the highest Context Bleeding rate (75.6%). This pattern is consistent with strong in-context conditioning anchoring generation to prior frames, which may make previously established entities difficult to remove.

Amnesia (Type II) in Cascaded Pipelines. VGoT has the highest Amnesia rate (82.1%), consistent with a planning–rendering bottleneck in which an entity omitted from the initial text-to-image layout is difficult for the subsequent motion module to introduce.

Spurious Transitions (Type III) via Noisy Exploration. SD records the highest spurious-transition rate (51.4%). One possible explanation is that CSA feature sharing permits greater state variation, occasionally producing unprompted changes while also avoiding the frozen narratives observed in more strongly conditioned systems. This interpretation is correlational rather than causal.

Reversal (Type IV) as a Universal Blind Spot. Although relatively rare (7.6–13.9%), Reversals—where the observed transition is opposite to the prescription—occur in every evaluated system. As compounded errors with $|E_{t,e}| = 2$, they indicate a recurring shared difficulty in distinguishing the direction of entity entrances and exits.

6.5 The Consistency–Dynamics Trade-off

Figure 3 visualizes the **consistency–dynamics association**: the four evaluated systems lie on a strict inverse ordering ($\rho_s = -1.0$). This is consistent with a tension between cross-shot coherence and localized entity control, but $n = 4$ is insufficient to establish a universal or causal relationship.

Table 6: Failure distribution (% of videos with ≥ 1 failure).

Failure Type	SD	Echo	VGoT	VIC
I: Context Bleeding(↓)	64.1	66.9	73.0	75.6
II: Amnesia(↓)	69.1	73.1	82.1	80.6
III: Spurious Trans(↓)	51.4	42.2	25.2	20.2
IV: Reversal(↓)	13.0	13.9	7.6	11.5
No failure↑	1.5	0.4	0.3	0.2

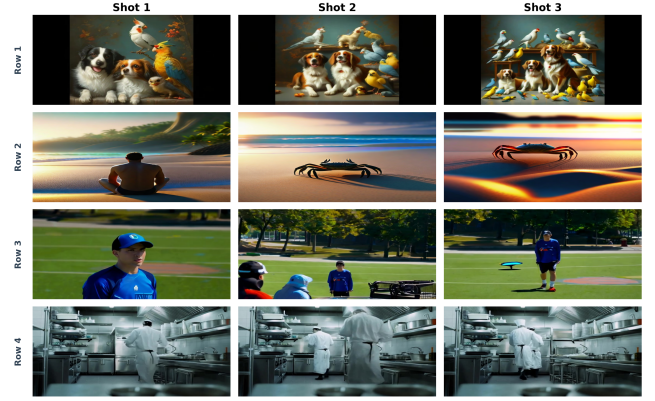


Figure 2: Qualitative examples of the four failure types on real generated frames. Each row corresponds to one failure type; see Table 7 for full descriptions.

Table 7: Failure type details for Fig. 2. *Prompt* shows the intended pattern; *Observed* shows the actual outcome, with deviations in bold.

Row	Failure Type	Entities	Prompt	Observed
1	Context Bleeding (I)	A: cat, B: dog, C: parrot	$ABC \rightarrow AB \rightarrow A$	$AB \rightarrow AB \rightarrow AB$
2	Amnesia (II)	A: swimmer, B: crab	$A \rightarrow B \rightarrow AB$	$A \rightarrow B \rightarrow B$
3	Spurious (III)	A: frisbee player, B: spectator	$A \rightarrow B \rightarrow AB$	$A \rightarrow AB \rightarrow A$
4	Reversal (IV)	A: cook, B: dishwasher	$AB \rightarrow A \rightarrow B$	$A \rightarrow AB \rightarrow A$

6.6 Human Evaluation

To validate our automated metrics against human perception, 15 evaluators (graduate students in CS and media studies) assessed 40 randomly sampled scenarios. Evaluators conducted pairwise comparisons of generated videos alongside the original per-shot text prompts, rating each model’s *Narrative Compliance* on a 5-point Likert scale and selecting a preference (A win / B win / MutualFail / BothGood). Each evaluator completed 120 comparisons; left/right presentation was randomized (see Appendix J for full protocol).

The human ranking—SD > Echo > VIC > VGoT—strongly agrees with the automated PTM ranking ($\rho_s = 0.80$), with only VIC and VGoT swapped in the bottom tier. In decisive comparisons, SD achieves a 57.7% win rate, leading all head-to-head matchups. Bradley–Terry analysis [3] confirms a clear tier separation: SD

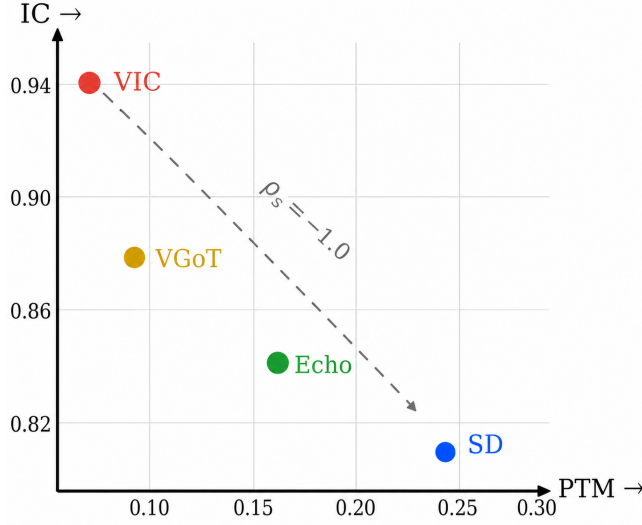


Figure 3: Consistency–Dynamics association in the evaluated set. The four systems exhibit an inverse IC–PTM ordering ($\rho_s = -1.0$, $n = 4$): VIC has the highest IC and lowest PTM, while SD has the highest PTM and lowest IC.

(strength 1.24) vs. the remaining models (0.80–1.02). Critically, 62.4% of all comparisons were labeled *MutualFail*—independently corroborating our automated finding that current models frequently fail to execute prescribed narrative transitions.

6.7 Discussion & Limitations

Detector Reliability vs. Generation Failure. The raw presence match against S^* (52% TPR) conflates detector errors with generation failures. We manually inspected 100 cells with $S_{t,e}^* = 1$ but $S_{obs}[t, e] = 0$: 65% were genuine generation failures, 19% were ambiguous distortions, and 16% were VLM errors. Among the 81 non-ambiguous cases, 65 (80.2%) therefore reflected missing generated entities rather than missed detections. Absolute PTM/EPA values nevertheless remain dependent on the specified detector.

Detector Robustness and Discriminative Validity. Replacing Qwen2-VL with LLaVA-1.6-7B [22] changes PTM from (.243, .167, .094, .070) to (.238, .124, .094, .060) for (SD, Echo, VGoT, VIC), but preserves the ordering exactly ($\rho_s = 1.00$). Together with the multi-frame results, this indicates that the comparative ordering is not an artifact of one detector or sampling choice. Qwen2-VL-7B also provides a better balance of TPR and TNR than the Grounding DINO and 2B configurations evaluated in Appendix K, while the Kruskal–Wallis test ($H = 302.7$, $p < 10^{-65}$), bootstrap confidence intervals, and pairwise significance tests support discrimination under the primary protocol.

Metric Decomposition. A naive Frobenius norm $\|E\|_F$ achieves significance in only 1 of 6 pairwise comparisons ($\alpha = 0.05$), because static hold cells outnumber active transitions by 2–4 $\times$, drowning the narrative signal in detector noise. The sparsity-aware decomposition into PTM and EPA is thus an empirical necessity, not merely a theoretical choice.

Five-Shot Stress Test. The main benchmark uses $T = 3$ and $N \leq 3$; theoretical extensibility is not long-form empirical validation. We therefore test 100 five-shot scenarios per system (400 videos). PTM/EPA are SD .403/.719, VGoT .215/.679, Echo .169/.649, and VIC .073/.658. SD remains strongest and VIC weakest, but VGoT overtakes Echo in the middle tier. We present this as a stress test showing persistent longer-horizon difficulty, not exhaustive long-form coverage.

Independent Querying Ambiguity. Our STEM pipeline evaluates entity presence through independent binary queries. Visually similar entities can induce compositional hallucinations, causing an absent entity to be reported as present. MEMS-1K mitigates this through the CLIP-similarity threshold (≤ 0.45 , §5), at the cost of underrepresenting similar-character narratives. Stronger joint-reasoning detectors are needed to remove this construction trade-off.

7 Conclusion

We introduced SPAN, an algebraic representation of entity-level multi-shot transitions, and STEM, a diagnostic pipeline based on sparsity-aware PTM/EPA metrics. The completeness theorem characterizes the canonical matrix space, while MEMS-1K samples seven interpretable topologies rather than exhaustively enumerating all feasible narratives. Across four representative systems, STEM detects significant capability gaps ($H = 302.7$, $p < 10^{-65}$), fewer than 1.5% error-free videos, and a four-type failure taxonomy. Human evaluation ($\rho_s = 0.80$), multi-frame aggregation, and an alternative detector support the comparative conclusions; the five-shot stress test further shows persistent longer-horizon difficulty. We release the prompts, evaluation code, and 4,000 generated videos.

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